Arduino — RFID Access Control System

A Hands-On Electronics Lesson for Young Makers

Lesson Overview

What is Arduino?

Arduino is a tiny programmable computer that controls things like lights, buzzers, and motors. Think of it as a robot brain — you write instructions and it brings your project to life.

What is RFID?

RFID stands for Radio Frequency Identification. Special cards carry a tiny chip that sends a unique secret code to a reader. Tap your card and the reader instantly recognises you — science that feels like magic!

How Does RFID Access Control Work?

Step 1	**Scan** Hold your RFID card near the reader module.
Step 2	**Check** Arduino compares the card's ID against the allowed list in the code.
Step 3	**Access** Match found → green LED lights up, buzzer beeps, door unlocks!
Step 4	**Deny** No match → nothing happens and the door stays locked.

Key Components

Arduino Uno	The brain of the project — a microcontroller board you program using the Arduino IDE.

RC522 RFID Reader	Reads the unique ID from RFID cards/key-fobs using radio waves (13.56 MHz).
RFID Card / Tag	Contains a tiny chip that broadcasts its unique ID when near the reader.
Green & Red LEDs	Visual indicators — green = access granted, red = access denied.
Buzzer	Provides audio feedback: a happy beep for granted access.
Breadboard & Jumper Wires	Used to connect all components without soldering.
Resistors	Protect the LEDs from receiving too much current and burning out.

Wiring Diagram — Arduino Pin Connections

Component	Pin / Leg	Arduino Pin
	SDA	Pin 10
	SCK	Pin 13
	MOSI	Pin 11
RFID Reader (RC522)	MISO	Pin 12
	RST	Pin 9
	GND	GND
	3.3V	3.3V
Buzzer	Positive (+)	Pin 8
Buzzer	Negative (–)	GND
Green LED	Anode (+)	Pin 7 (via resistor)
Red LED	Anode (+)	Pin 6 (via resistor)
	Cathode (–)	GND

Sample Arduino Code

The sketch below is the core logic for the RFID access control system:

```
#include
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#define SS_PIN 10
#define RST_PIN 9
#define GREEN_LED 7
#define RED_LED 6
#define BUZZER 8
MFRC522 rfid(SS_PIN, RST_PIN);
// Add your allowed card UIDs here
String allowedUID = "A3 F2 11 4B";
void setup() {
Serial.begin(9600);
SPI.begin();
rfid.PCD_Init();
pinMode(GREEN_LED, OUTPUT);
pinMode(RED_LED, OUTPUT);
pinMode(BUZZER, OUTPUT);
}
```

```
void loop() {
if (!rfid.PICC_IsNewCardPresent()) return;
if (!rfid.PICC_ReadCardSerial()) return;
String uid = "";
for (byte i = 0; i < rfid.uid.size; i++) {
uid += String(rfid.uid.uidByte[i], HEX) + " ";
}
uid.trim();
uid.toUpperCase();
if (uid == allowedUID) {
Serial.println("Access GRANTED");
digitalWrite(GREEN_LED, HIGH);
tone(BUZZER, 1000, 300);
delay(2000);
digitalWrite(GREEN_LED, LOW);
} else {
Serial.println("Access DENIED");
digitalWrite(RED_LED, HIGH);
delay(1000);
digitalWrite(RED_LED, LOW);
}
rfid.PICC_HaltA();
}
```

Troubleshooting Tips

■ **Wires loose?**	Check every jumper wire is firmly seated in the breadboard and Arduino pins.
■ **Card not recognised?**	Open the Serial Monitor in Arduino IDE to print the scanned UID, then copy it into your allowedUID variable exactly.
■ **LED not lighting up?**	Ensure you have a current-limiting resistor (220Ω–330Ω) in series with each LED.
■ **Buzzer silent?**	Confirm the positive leg is connected to the correct pin and negative to GND.
■ **Upload error?**	Select the correct board (Arduino Uno) and COM port in Tools menu of the Arduino IDE.
■ **Nothing works?**	Unplug the USB cable, wait 5 seconds, and reconnect. Then try uploading again.

■ Safety Tips

✓ Always unplug your Arduino before changing any wiring or connections.

✓ Never touch components when the circuit is powered on — you could get a shock.

✓ Always work with adult supervision when using electronics and tools.

✓ Keep components away from water or conductive surfaces.

What You Learned Today

1. How Arduino and RFID work together to create a smart access control system.
2. Basic electronics wiring using breadboards, jumper wires, and components.
3. Programming concepts: reading card IDs and controlling LEDs and buzzers.
4. Real-world applications: schools, offices, and libraries all use RFID access control.
5. Safety practices when working with electronics.

Keep Exploring & Keep Building! ■